

20250224EB_MN (GC-MS)

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- 2025年2月24日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2025年2月26日 星期三
- T2 -> T3
- 2025年2月28日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2025年3月3日 星期一
- T4 -> T5
- 2025年3月5日 星期三
- T5 -> T6
- 2025年3月7日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2025年3月10日 星期一
- T6 -> T7
Dissociation
 - Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
 - Remove the PLO and wash the plate with PBS once
 - Coat the plate with laminine for 1hr.
 - Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
 - Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
 - Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
 - Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
 - Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

***EBs are not much this time

Table3							^
	A	B	C	D	E	F	
1	conc. (M/mL)	survival	total (4mL)	M/well (into 6*6-well)	volume for 1.2M (uL)	the rest for 24-well*8 (K/well)	
2	2.25	100	9	1.2	533	225	

Well1				^
	1	2	3	
A	2.5mM BP T8	2.5 mM C13-glu in Neurobasal-A		
B	0.15mM C12-glu in neurobasal-A+ Na pyruvate	0.15 mM C13-glu in Neurobasal-A		

plate B for astrocyte co-culture							^
	1	2	3	4	5	6	
A							
B							
C							
D							

2025年3月12日 星期三

T7-> T7

2025年3月13日 星期四

T7 -> BP T8

2025年3月15日 星期六

700uL BP T8 -> 800 uL BP T8

2025年3月17日 星期一

700uL BP T8 -> 800 uL BP T8

2025年3月18日 星期二

Change media with heavy glucose at around 2:30pm

2025年3月19日 星期三

GC-MS extraction with Zixuan

2025年3月22日 星期六

Imaging: plate B (failed)

new vLH92 didn't work. only find <5 cells with signal in 4 wells.

2025年3月26日 星期三

GC result discussion with Zixuan

- Pyruvate amount is very high in non-labeled samples -> because I used the BP and hypo media in which I added Na Pyruvate.
- Alanine should be pretty similar to pyruvate
- Alanine pool big? Or just slower synthesized?
- Glycine is really abundance in cells hard to label
- Valine -> essential aa -> should be really similar
- Succinate is low in cells
- M4 is a lot for our experiment. (M4 will be appear in the second round)
- BP has really low glycine, serine, but high alanine
- Essential aa should not see label.
- Aspartate looks great
- 5-10% difference in fractional labeling is large. Therefore, replicate is important
- Glutamate and citrate have high labeling
- Pyruvate, lactate, alanine is similar in other cell types
- For next experiment:
 - a. Do not use the media with sodium pyruvate
 - b. Use 1-2 wells with coating & media without cell
 - c. Ask for brainphys and stemflex compositions